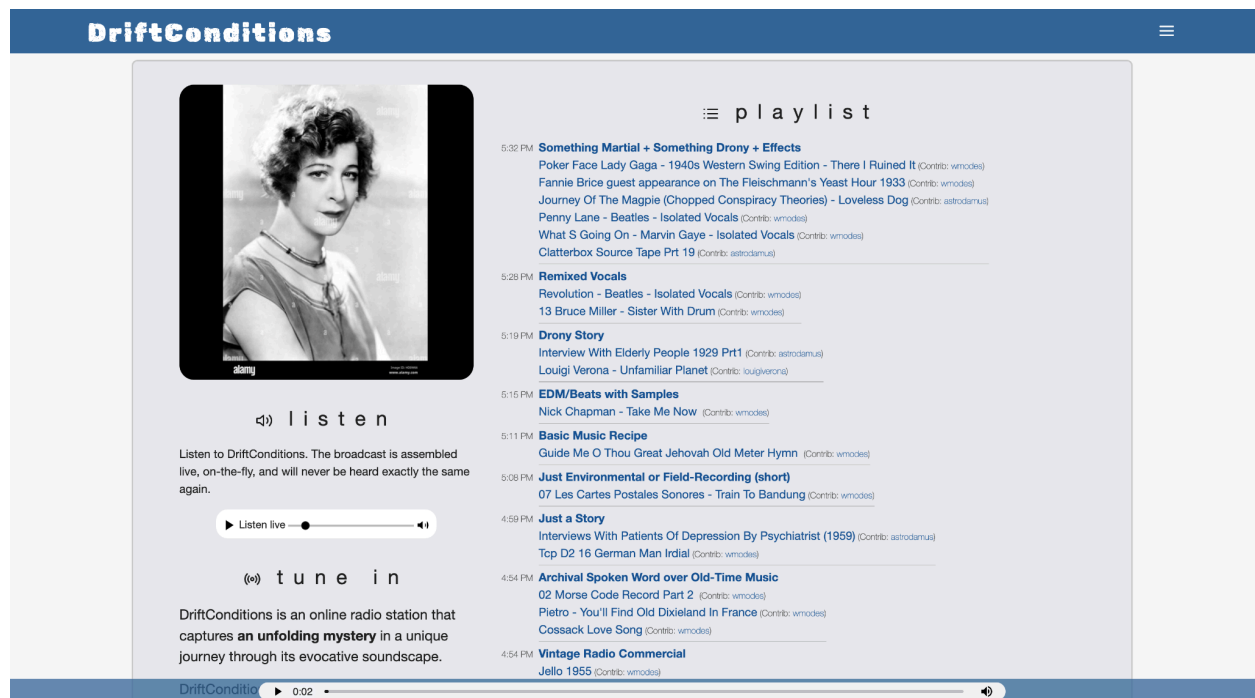


DriftConditions: A Generative Audio Streaming Platform

Project Documentation



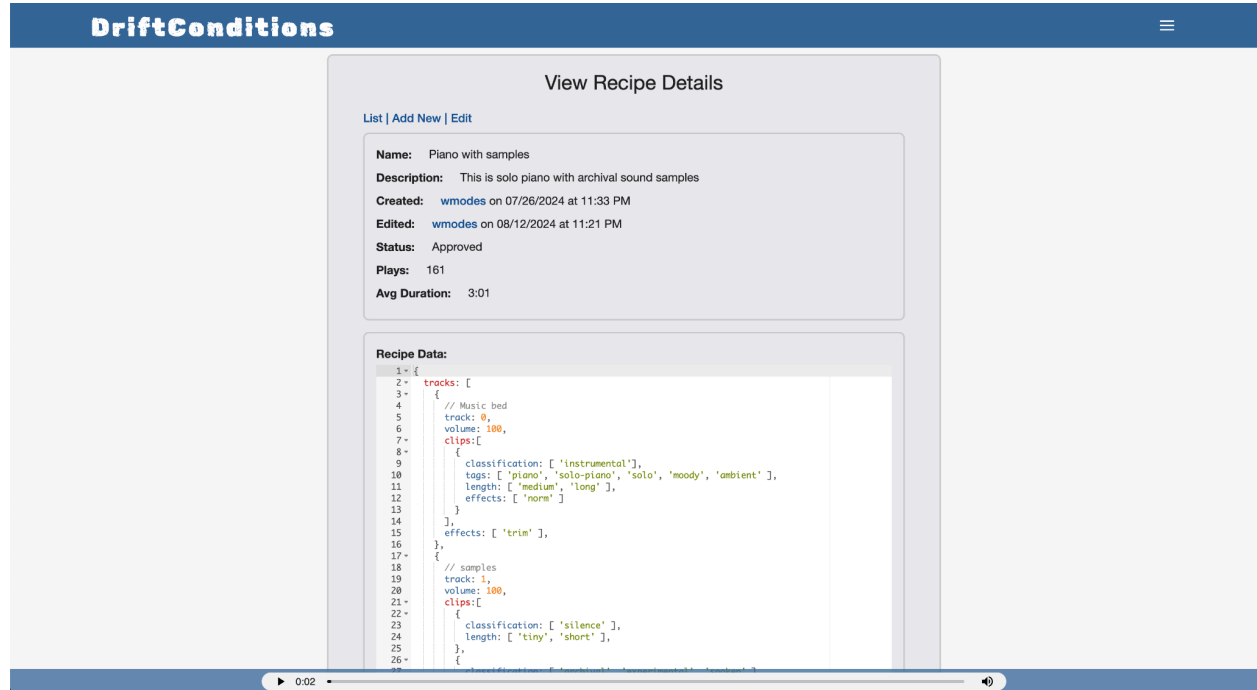
The Work

DriftConditions is a continuously running generative sound system and creative research project. Drawing on stochastic processes, procedural composition, and a live community-fed audio archive, it produces emergent configurations rather than fixed compositions. The work investigates how unpredictability, drift, and constraint-based authorship can generate shifting narrative textures and charged listening states in which voices, memory, and environment intermingle.

For the listener, the result is an uncanny online audio source that captures the chaos and serendipity of late-night radio tuning: overlapping fragmented stories, ambient sounds, and mysterious crosstalk weaving an immersive and unpredictable sonic field. Rather than presenting a finished composition, the system continually assembles new relations among contributed materials, inviting introspection through impermanence, interference, and surprise.

The result is a living, participatory archive that produces endlessly shifting sonic experiences, where voices, memory, and environment intermingle in unpredictable ways.

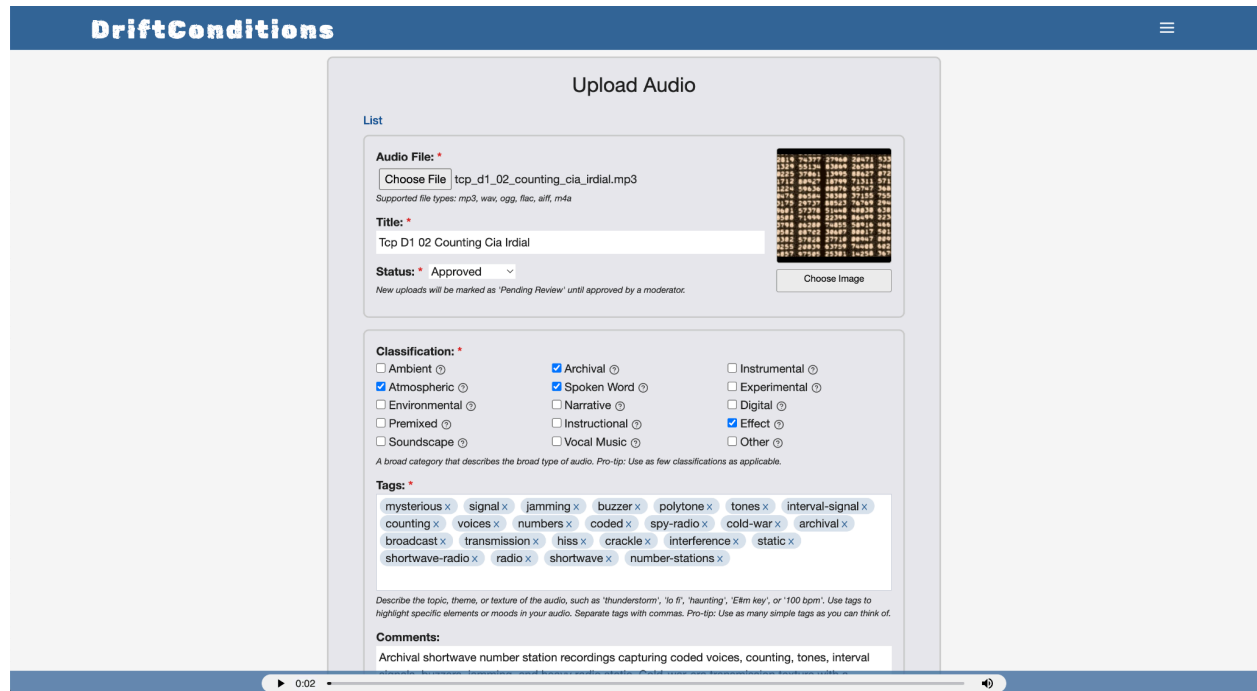
The stream is live and public at driftconditions.org.



Community Participation

The station relies on community audio contributions. Contributors upload audio clips — music, field recordings, spoken word, ambient sound — through a web interface. Editors and moderators review and tag everything. A graduated permission system governs access: User → Contributor → Editor → Mod → Admin. Role changes trigger an email notification to the user and adjust their digest frequency automatically.

This contributor model is not just a platform feature but a methodological commitment. My broader practice has long centered participatory systems in which the work is shaped by the people who enter it, contribute to it, and leave traces within it. In *DriftConditions*, that means the archive is not simply author-curated from above but community-fed: the sound world emerges from distributed contributions, collective selection, and the procedural relations the system builds among them. That matters because it shifts the project from a private compositional engine toward a participatory archive in which authorship, memory, and listening are shared and continually recomposed.



Scope of Work (2025–2026)

The *DriftConditions* platform was already operational and proven in exhibition in 2024. It now exists as a publicly released, open-source, DOI-archived generative media platform with a portable architecture, a functioning contributor ecosystem, and a documented path for adaptation into new research contexts.

Over the course of Fall 2025 and Spring 2026, the platform was developed across four major areas:

Platform portability. Brand identity was previously scattered across hardcoded JSX strings, CSS variables, a Tailwind config, and inline email templates in route handlers. This made the platform fragile to rebrand and impossible to adapt. All site identity, copy, and visual branding were extracted into a reusable abstraction layer. A new deployment now requires only editing `brand.js` and copy files to produce a fully re-skinned instance. This is the key architectural pivot of the year: *DriftConditions* becomes not only a single artwork, but a reusable foundation for future procedural archive projects.

Authentication and identity. The system now supports OAuth 2.0 sign-in via Google, GitHub, and Discord, linking new OAuth identities to existing profiles where emails match and maintaining consistent role-based access control across every authentication method.

Compositional engine. The MixEngine is the core of what *DriftConditions* does — everything the listener hears flows through it. Several known bugs degraded the quality of that experience: clips outside the requested length range were occasionally selected, volume balance between

tracks was sometimes wrong, and certain recipes misbehaved. These were fixed. New capabilities were added, including fadeout, looping, repeat, ducking, recipe-duration weighting, and usage-weighted clip selection that prefers less-used clips over time.

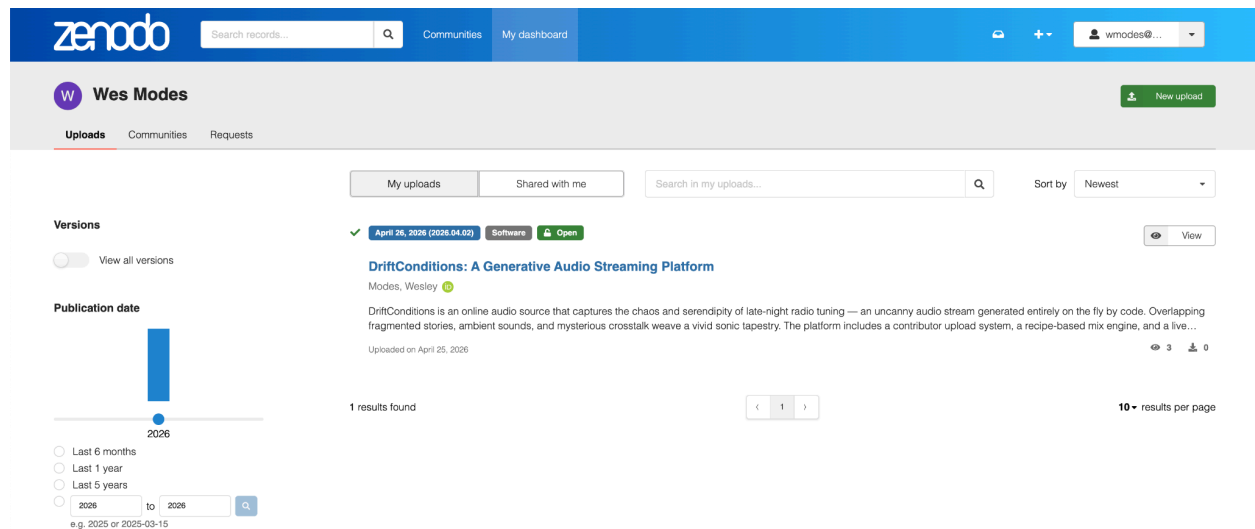
Contributor infrastructure and interface. Contributors previously received little feedback when their audio was approved, rejected, or played. A transactional email system was built — welcome messages, password reset, contributor digests, moderation notices, and role change notifications — using templated emails sent on a per-user schedule. At the same time, the editorial backend was overhauled to enhance consistency and clarity for listeners and editors, streamline contributor workflow, and introduce search and usage tracking within the audio library.

DriftConditions										
Audio List										
Search title, tags, comments... Search										
All Review Approved Disapproved Trash										
2390 items 14 15 16 17 18 page 1 of 160 19 20 21										
ID	Title	Author /Date	Plays	Duration	Status	Type	Tags			
2491	Tcp D4 43 M3b Indial	Upload: wmodes on 04/26/2026 at 5:45 PM	—	0:49	Approved	archival, atmospheric, spoken, effect	mysterious, signal, jamming, buzzer, polytone, tones, interval-signal, counting, voices, numbers, coded, spy-radio, cold-war, archival, broadcast, transmission, hiss, crackle, interference, static, shortwave-radio, radio, shortwave, number-stations, archive-org			
2490	Tcp D4 42 M3a Indial	Upload: wmodes on 04/26/2026 at 5:45 PM	—	0:22	Approved	archival, atmospheric, spoken, effect	mysterious, signal, jamming, buzzer, polytone, tones, interval-signal, counting, voices, numbers, coded, spy-radio, cold-war, archival, broadcast, transmission, hiss, crackle, interference, static, shortwave-radio, radio, shortwave, number-stations, archive-org			
2489	Tcp D4 41 M3 Indial	Upload: wmodes on 04/26/2026 at 5:45 PM	—	0:15	Approved	archival, atmospheric, spoken, effect	mysterious, signal, jamming, buzzer, polytone, tones, interval-signal, counting, voices, numbers, coded, spy-radio, cold-war, archival, broadcast, transmission, hiss, crackle, interference, static, shortwave-radio, radio, shortwave, number-stations, archive-org			
2488	Tcp D4 40 M3 Indial	Upload: wmodes on 04/26/2026 at 5:45 PM	—	0:15	Approved	archival, atmospheric, spoken, effect	mysterious, signal, jamming, buzzer, polytone, tones, interval-signal, counting, voices, numbers, coded, spy-radio, cold-war, archival, broadcast, transmission, hiss, crackle, interference, static, shortwave-radio, radio, shortwave, number-stations, archive-org			
2487	Tcp D4 39 M3 Indial	Upload: wmodes on 04/26/2026 at 5:45 PM	—	0:17	Approved	archival, atmospheric, spoken, effect	mysterious, signal, jamming, buzzer, polytone, tones, interval-signal, counting, voices, numbers, coded, spy-radio, cold-war, archival, broadcast, transmission, hiss, crackle, interference, static, shortwave-radio, radio, shortwave, number-stations, archive-org			
2486	Tcp D4 38 M3 Indial	Upload: wmodes on 04/26/2026 at 5:45 PM	—	0:27	Approved	archival, atmospheric, spoken, effect	mysterious, signal, jamming, buzzer, polytone, tones, interval-signal, counting, voices, numbers, coded, spy-radio, cold-war, archival, broadcast, transmission, hiss, crackle, interference, static, shortwave-radio, radio, shortwave, number-stations, archive-org			
2485	Tcp D4 37 M3 Indial	Upload: wmodes on 04/26/2026 at 5:45 PM	—	0:19	Approved	archival, atmospheric, spoken, effect	mysterious, signal, jamming, buzzer, polytone, tones, interval-signal, counting, voices, numbers, coded, spy-radio, cold-war, archival, broadcast, transmission, hiss, crackle, interference, static, shortwave-radio, radio, shortwave, number-stations, archive-org			
2484	Tcp D4 36 M3 Indial	Upload: wmodes on 04/26/2026 at 5:45 PM	—	0:32	Approved	archival, atmospheric, spoken, effect	mysterious, signal, jamming, buzzer, polytone, tones, interval-signal, counting, voices, numbers, coded, spy-radio, cold-war, archival, broadcast, transmission, hiss, crackle, interference, static, shortwave-radio, radio, shortwave, number-stations, archive-org			
2483	Tcp D4 35 M2 Indial	Upload: wmodes on 04/26/2026 at 5:45 PM	—	3:09	Approved	archival, atmospheric, spoken, effect	mysterious, signal, jamming, buzzer, polytone, tones, interval-signal, counting, voices, numbers, coded, spy-radio, cold-war, archival, broadcast, transmission, hiss, crackle, interference, static, shortwave-radio, radio, shortwave, number-stations, archive-org			
2482	Tcp D4 34 M1b Indial	Upload: wmodes on 04/26/2026 at 5:45 PM	—	4:47	Approved	archival, atmospheric, spoken, effect	mysterious, signal, jamming, buzzer, polytone, tones, interval-signal, counting, voices, numbers, coded, spy-radio, cold-war, archival, broadcast, transmission, hiss, crackle, interference, static, shortwave-radio, radio, shortwave, number-stations, archive-org			
2481	Tcp D4 33 M1 Indial	Upload: wmodes on 04/26/2026 at 5:45 PM	—	0:51	Approved	archival, atmospheric, spoken, effect	mysterious, signal, jamming, buzzer, polytone, tones, interval-signal, counting, voices, numbers, coded, spy-radio, cold-war, archival, broadcast, transmission, hiss, crackle, interference, static, shortwave-radio, radio, shortwave, number-stations, archive-org			

Release and Archival

DriftConditions **v2026.04.02** was publicly released and archived in Zenodo in April 2026 as open-source software under the MIT license.

- DOI: **10.5281/zenodo.19774081**
- Repository: github.com/wmodes/driftconditions
- License: MIT
- Scale: 570 commits, ~20,000 lines of original code, four interconnected services



Research Context

Two ongoing bodies of work form the foundation of this project: *DriftConditions*, a procedural generative sound system, and *A Secret History of American River People*, a long-term participatory archive of river communities encompassing 175 oral histories across American river systems. The work described here lives at their intersection.

People's River History / DriftConditions asks: How can oral histories and fieldwork recordings be transformed into procedural soundscapes that invite listeners into immersive states where past and future rivers can be re-envisioned? The intellectual contribution lies in formalizing two artistic strategies — drift, inspired by the *dérive* and the meandering of rivers, and listening as method — into a reproducible framework for participatory archives. The project shows how procedural media can turn an archive into an experience rather than an inventory.

The platform portability work completed this year is what makes that next phase possible. By separating *DriftConditions*' underlying engine from its fixed site identity and surface copy, the project can now serve a new body of archival content without rebuilding the system from scratch. In research terms, this is the architectural move that transforms the project from a single generative artwork into a reusable framework for procedural participatory archives.

This year's work also produced a strategic pivot from building features to building research capacity — codifying standards, reducing coordination risk, and establishing a sustainable new media working group model with documented architectural standards and collaboration protocols.

This project was supported by a CDRI Research Project grant from DAAP (\$3,500, Fall 2025), "Building Capacity for Procedural Participatory Archives." It will generate the working prototype to anchor external funding applications, including NEA Media Arts Projects, NEH Digital Humanities Advancement Grant, and Ohio Humanities Major Grant.